

Abstract- Border surveillance is major factor in national security, yet the existing system having some of limitations like limited coverage and delayed response [10]. This paper presents smart border surveillance system using ESP-32 with swarm nodes integrated with multiple sensors like radar, Time-of-flight, GPS, and camera module [7],[11]. This system gives real time data of environment like detects motion, captures images, and verifies intrusions using sensor fusion and lightweight object detection and others [8],[18]. Servo and motors enable the mobile capturing of data and gps location updates. This system provides reliability and scalability and it is mainly cost effective, low power consumption and suitable for deployment in areas having low human coverage.

Keywords--Border Surveillance, ESP32, Swarm Intelligence, Internet of Things (IoT), Multi-Sensor Fusion, Radar Sensor, Time-of-Flight (ToF), ESP32-CAM, Object Detection, YOLO, GPS Tracking, Embedded Systems, Smart Monitoring, Edge Computing, Wireless Sensor Networks.

I.INTRODUCTION

Border surveillance is one of the major parts of national security aimed at protecting the nation from the enemy or the threats through human or robotic attack systems and major from the terror groups [10]. Traditional surveillance methods such as manual patrolling, fixed camera installations are used from decades are old technique in safety systems [14]. These systems have been facing several loopholes and system failures, operational cost energy consumption, maintenance in remote areas are very difficult and dangerous.

With increase in IoT and embedded field and systems there is growing shift to IoT, wireless communication and automated systems [4],[14]. Modern equipments are mainly try to reduce human involvement and dependency systems having with high accuracy and response time. By applying this thought, wide area coverage and monitoring using multiple interconnected nodes has emerged as a major advancement in the system [19]. Instead of relying on a single centralized system, multiple low-cost devices can be deployed across a wide area to provide continuous and combined real time monitoring.

In this paper the major focus is in smart border surveillance system based on ESP32 microcontroller nodes operating in a swarm-like architecture [7],[19]. Each node is equipped with multiple sensors, including radar for motion detection, Time-of-Flight (ToF) sensors for distance measurement, and camera modules for visual monitoring [11],[12],[13]. The combination of these allows to work in multilevel processing and receiving real time coverage and verification of the area and environment surrounding reducing false alarm, GPS system to provide present location and radar coverage system to identify the environment system.

This system also has motor system using servo and N20 encoder motors for identifying camera positions, object detection to identify humans and capture suspicious objects [8],[18]. This system has decentralized by having nodes which can process data locally and communicate using nearby nodes forming communicative nodes, this system increases scalability and coverage for large areas [9],[19].

Totally this system aims in providing a cost effective, scalable combined processing system using embedded technology and most growing area of interest in national security and surveillance combining decentralized communication system, covering major system limitations present in traditional existing methods and offers one of the effective systems using modern technology for security systems[20].

Literature Survey

In recent years, many researchers have worked on improving border surveillance by using IoT and embedded systems [4], [14]. These systems make it easier to monitor large areas and collect data in real time [9], [17]. Some studies have focused on ESP32-based designs and wireless sensor networks for remote monitoring [7], [10]. Deep learning methods, especially for object detection, have also helped in identifying intrusions more accurately [8], [18]. To make systems more reliable, researchers often combine sensors like radar, ToF, and cameras to reduce false alarms [11], [12], [13]. There is also growing interest in swarm-based systems for better coverage [5], [19], although issues like cost, coordination, and centralized control still remain [20].

Work Methodology

This project smart border surveillance system operates using a combination of multi-sensor detection, edge processing, and swarm-based communication [9], [11]. The major principle behind the working of the system is to ensure real time monitoring, accurate intrusion detection and effective response to human intervention is observed.

The system initially used radar system to continuously scan the area and detect motion [12]. Radar is preferred since it can work effectively in low visibility and darkness, fog. If the motion is detected the ESP32 microcontroller activates the ESP32-CAM module, which captures images of the monitored region [7]. These signals are processed to detect the object whether it is human or any suspicious entity [8].

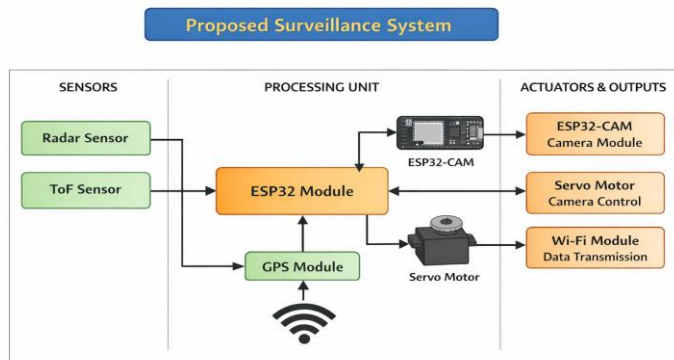
To improve accuracy and avoid false alarms, the system uses a Time-of-Flight (ToF) sensor [13]. The ToF sensor measures the distance between the object and the surveillance node, helping confirm whether the detected motion is relevant. This combination of radar and distance sensing significantly increases detection reliability compared to single-sensor systems.

The system has servo motor for movement of camera in adjusting the angle. When an object is detected, the servo motor rotates the camera to track the movement, ensuring continuous monitoring. N20 encoder motor is used for deployment in remote areas and for implementing as mobile unit with continuous movement around the object.

GPS module is integrated into each node to provide real-time location data. Whenever an intrusion is detected, the exact coordinates are recorded and transmitted along with the alert. This helps in quick identification of the intrusion location.

All the processed data, including sensor readings, captured images, and GPS coordinates, are transmitted via Wi-Fi to a monitoring system. The system can display real-time alerts and status updates for effective decision-making.

Block diagram



The sensing unit includes radar and ToF sensors, which detect motion and measure distance. The processing unit is the ESP32, which controls all operations and processes the data. The ESP32-CAM provides visual input, while the servo motor controls camera movement. The GPS module adds location tracking capability. The communication uses WIFI to send data to monitoring system.

- A. Continuous Monitoring - Using radar the environment is scanned continuously, to detect movement in area with low visibility like darkness or fog. The system remains active to detect suspicious activity.
- B. Image Capture and Detection- The ESP32-Cam is used to detect the motion and capture the image. The captured images are analysed using object detection method and classified the object as human or potential threat.
- C. Distance Verification- The Time-of-Flight sensor is used to measure the distance between the system and object detected. This reduces the false detection and false alarms due to environmental movements.
- D. Tracking and Localization- The servo adjusts the camera angle to track the movement of object. GPS module records the exact location of the event occurring for effective tracking and detection.
- E. Data Transmission and coordination- The processed data and captured images are transmitted via Wi-Fi to the base system. The swarm network communicates with multiple nodes, increasing the coverage and reliable working across the large surveillance area.

Multi-Sensor Analysis

This proposed surveillance system mainly depends on how precise the multiple sensors work together. Instead of depending on a single data source, this combines number of inputs from different sensors to improve the accuracy and overall performance of the system. This multi-sensor system ensures the system working under challenging environmental conditions such as dark area, fog or obstacles.

A. System Integration and Data collection

The system is designed to work under the control of ESP32 microcontroller controlling all the sensors [7]. The real time monitoring of environment data is carried by different sensors each sensor plays a specific role in real time data collection. The radar system continuously scans the area for motion detection [12], acting as primary data collection system. Once the motion is

detected ESP32 microcontroller activates the ESP32-CAM to capture the images and sent to processing.

Parallely the ToF sensor collects the distance information about detected object. Using this data the we can determine whether the object is in relevant range. The GPS module detects the geographical location and ensures the precise mapping of the object for detecting the exact location of object.

All these data streams motion detection, visual input, distance measurement, location are gathered by ESP32. The collection of data is continuous and synchronized with no critical event is missed. The combined data from different sensors makes clear understanding rather than single sensor system.

B. Sensor Integration

Sensor integration is major part in the system, where different data from multiple sensors is combined to make decisions. Each sensor has its own limitation and strengths. Example radar is super effective in detecting the motion but cannot identify the type of object. Camera based systems can identify the object but may struggle in poor lightning conditions.

To tackle these limitations system integrates output of different sensors. When radar detects motion, it triggers the camera to capture image. The image is analysed to identify the object. The ToF sensor verifies the environment noise like leaves movement, animals.

Additionally, servo motor enhances sensor integration by adjusting the camera direction based on the movement. This makes the system can dynamically focus on areas of interest, this improves surveillance coverage.

C. Data Analysis

The data is collected and integrated, it is analysed to make decisions. The ESP32 processes the sensor data locally, enabling faster response times compared to systems that rely completely on cloud processing. This procedure involves checking motion signals, verify distance and interpreting the visual data from the camera.

A lightweight object detection method is used to analyse captured images and identify potential threats. The system compares the outputs from different sensors to confirm whether the detected activity is significant. If motion is detected but no object is identified in the image, the system may ignore the event as a false alarm.

When all conditions are satisfied, the system classifies the event as a valid intrusion. The GPS coordinates are then attached to the alert, and the information is transmitted to the monitoring system. This helps in quick and accurate decision making.

Overall, the multi-sensor data analysis improves decision making by combining different types of information. It ensures the system is accurate and reliable, making it suitable for real world application.

Feedback Analysis

The performance of the proposed surveillance system was evaluating based on the result of continuous testing under different environment conditions. By observing the system behaviour like

detection accuracy, response time and reliability the feedback was noted. It was observed that the integration of multiple sensors made the system very reliable and improved systems efficiency [11]. The detection of intrusions were very accurate compared to single sensor approach.

During testing, the radar sensor effectively detected motion even in low-light and partially obstructed conditions. Radar detection sometimes resulted in false triggers due to environmental disturbances. This issue was addressed by combining radar data with ToF sensor measurements and camera-based detection [13],[8], which helped verify the presence of actual objects.

The object detection module provided accurate identification in most cases, especially in well-lit conditions. The use of servo-based camera tracking further improved monitoring by ensuring that moving objects remained within the camera's field of view. GPS integration also proved reliable [20], providing precise location data for detected events.

Overall, the system demonstrated stable performance with very less false alarms and real time data collection and response. The multi sensor fusion and edge processing are effective in enhancing the real time surveillance, making the system suitable for practical deployment.

Conclusion

This work presents a smart border surveillance system built using ESP32-based nodes that operate together in a swarm-like manner [7],[19]. The main aim was to overcome the drawbacks of traditional systems, especially their dependence on manual monitoring and slower response. By using a combination of radar, ToF, and camera modules, the system is able to detect intrusions more reliably through cross-verification of sensor data.

One of the key advantages of the system is its use of local processing, which helps in making faster decisions without depending heavily on cloud services. The addition of GPS makes it easier to identify the exact location of detected events, which can be useful for quick action. Features like camera rotation using a servo motor and optional movement using motors also improve how effectively the area can be monitored [11], [13], [8].

During testing, the system performed consistently across different conditions, showing fewer false alarms and quicker responses. The use of multiple nodes working together also makes the system more flexible and scalable[20].

Overall, the design offers a practical and affordable solution for real-world surveillance needs. Further improvements can focus on better AI models, power efficiency, and scaling the system for larger deployments.

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